

RS002 Week 4 — English Worksheet

Topic: Parallel Lists + Loops — A Full Pokédex · **Course:** Pokédex App · **Time:** about 45 minutes

In the live lesson your three starters grew into a **full Pokédex of 12+ Pokémon** across four parallel lists — name, type, HP, attack. This worksheet is about *thinking through* that code and being able to **explain it in your own words**: how a **for loop** walks every Pokémon by index, and why the parallel lists must stay the same length.

Words to know: **parallel list**, **for loop**, **index**, **len**, **same length**

1 · Recall

From memory:

Write a list called `names` holding three Pokémon, then write the line that prints the second one.

2 · Predict 🧠

Read each block. Write what you think will happen — just from reading, no running.

```
names = ["피카츄", "이상해씨", "파이리"]
for i in range(len(names)):
    print(names[i])
```

`len(names)` is 3, so `range(len(names))` gives 0, 1, 2 . What three lines print?


```
names = ["피카츄", "이상해씨", "파이리"]
types = ["전기", "풀", "불"]

for i in range(len(names)):
    print(f"{names[i]} - {types[i]}")
```

The loop uses `i` in both lists. Write out every line it prints.


```
names = ["피카츄", "이상해씨", "파이리"]
hp = [35, 45]

for i in range(len(names)):
    print(f"{names[i]}: HP {hp[i]}")
```

`names` has 3 items but `hp` has only 2. What happens on the last loop? Why?

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it on paper, then explain the fix.

Bug A — This should print **every** name in the list. Right now it only ever prints the first one.

```
names = ["피카츄", "이상해씨", "파이리"]
for i in range(len(names)):
    print(names[0])
```

Hint: the loop variable `i` changes each time — use it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A new Pokémon was added to `names` but not to the other lists. When the loop reaches the new one, the program crashes.

```
names = ["피카츄", "이상해씨", "파이리", "꼬부기"]
types = ["전기", "풀", "불"]

for i in range(len(names)):
    print(f"{names[i]} - {types[i]}")
```

Hint: parallel lists must all be the **same length**.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should number the Pokédex starting at `#1`. Right now the first entry shows `#0`.

```
names = ["피카츄", "이상해씨", "파이리"]
for i in range(len(names)):
    print(f"#{i} {names[i]}")
```

Hint: the index starts at 0, but humans count from 1.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

Read this working Pokédex closely. You do not run it — you read it and answer.

```
names = ["피카츄", "이상해씨", "파이리", "꼬부기"]
types = ["전기", "풀", "불", "물"]
hp = [35, 45, 39, 44]
attack = [55, 49, 52, 48]

print(f"\n=== 도감 ===")
print(f"등록된 포켓몬: {len(names)}마리\n")

for i in range(len(names)):
    print(f"#{i+1} {names[i]} 타입:{types[i]} HP:{hp[i]} 공격:{attack[i]}")
```

There are four lists. What makes them "parallel" — how do they line up with each other?

`len(names)` is 4. What does `range(len(names))` give the loop, and how many times does the loop run?

Inside the loop, `i` is used in all four lists at once. On the third time through the loop, what is `i`, and which Pokémon's data gets printed?

Why does the line use `#{i+1}` instead of `#{i}` ?

If you added a 5th Pokémon to `names` but forgot the other three lists, what would break, and on which loop?

5 · Explain Your Lesson Code 🎥

Explain the Pokédex code **you wrote in today's live lesson** in a short phone video. You may show it running on screen. Teach it like the viewer is new. Try to use: **parallel list, for loop, index, len, same length**.

1. Show your full numbered Pokédex on screen.
2. Read your `for` loop out loud and say what `i` does each time through.
3. Point to your four lists and explain why they must all be the same length.
4. Show what changes when one Pokémon is added.

Write what you will say in your video. Plan it here before you record.

Submit ☒

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.